Student Intelligent Learning System (Mentor AI)

Comprehensive Functional Documentation
This document describes the major functionalities available to Students, Professors, and Administrators along with practical examples and workflows.

1. Platform Overview

Mentor AI is an AI-powered adaptive learning platform built for higher education institutions.
The platform combines curriculum management, AI learning assistance, adaptive practice, analytics, recommendations, and revision planning into a single ecosystem.

The platform supports three primary roles:
• Student
• Professor
• Admin

The objective is not only to provide content, but also to guide students throughout their semester using analytics-driven recommendations.

2. Student Module - Homepage

The homepage acts as the student's command center.

Features:
• Exam Readiness Score
• Pending Dues
• Today's Focus
• Study Streak
• Topics Covered
• Revision Queue
• AI Alerts

Example:
A student logs in and immediately sees:
- Exam Readiness: 68%
- Pending Dues: 4
- Today's Focus: Deadlocks
- Alert: 'System Calls has not been practiced for 6 days.'

This allows the student to know exactly what to study next without manually searching through subjects.

3. Student Module - Learning Page

The Learning Page is designed for concept understanding.

Capabilities:
• AI Notes Generation
• Explain Simpler
• Give Example
• Summarize
• Related Concepts
• Interview Cheat Sheets
• Save Notes
• Add To Revision

Example:
Topic: System Calls

Student actions:
1. Reads generated notes.
2. Uses Explain Simpler to understand kernel mode switching.
3. Uses Give Example to see real-world examples.
4. Saves notes for future revision.
5. Adds the topic to the revision queue.

The page acts as an AI tutor rather than a static notes repository.

4. Student Module - Practice Engine

The Practice Engine converts passive learning into active recall.

Practice Modes:
• Topic Based Practice
• Weakness Based Practice
• Revision Mode
• Exam Simulation

Features:
• Adaptive difficulty
• Session tracking
• Immediate feedback
• Mistake analysis

Example:
Student selects:
Operating Systems → System Calls

Session:
10 Questions
8 Correct
Accuracy = 80%

The engine stores:
• Sessions = 1
• Questions Attempted = 10
• Accuracy = 80%

These values are later used by the analytics engine.

5. Student Module - Analytics & Recommendations

The Analytics Engine calculates:

Accuracy:
Correct Answers / Questions Attempted

Confidence:
Confidence adjusts accuracy based on the number of sessions and questions.

Exposure:
Topics attempted / Total topics available.

Mastery Status:
• NOT_STARTED
• LEARNING
• WEAK
• STRONG

Recommendations Generated:
• Today's Focus
• Pending Dues
• Suggested Next
• Revision Queue
• AI Alerts

Example:
A student who answers 2 questions correctly receives 100% accuracy but low confidence.
This prevents false mastery.

6. Student Module - Roadmap

The Roadmap module creates a structured learning journey.

Features:
• Weekly plans
• Daily tasks
• Topic progression
• Completion tracking

Example:
Week 1:
- Process Management
- Threads
- CPU Scheduling

Week 2:
- Deadlocks
- Memory Management

Future weeks unlock based on progress.

7. Professor Module

The Professor Dashboard focuses on classroom management and analytics.

Features:
• Create classrooms
• Invite students
• Upload syllabus
• Upload resources
• Monitor class progress
• Identify weak topics
• Identify at-risk students

Example:
Professor uploads Operating Systems syllabus.
The AI extracts:
- Units
- Topics
- Subtopics

Professor can then monitor:
- Average accuracy
- Average confidence

- Weak topics across the class
- Student participation

8. Admin Module

The Admin Dashboard focuses on platform management.

Features:
• Total students
• Total professors
• Total classrooms
• Daily active users
• Recent activity
• System health

Example:
Admin can quickly determine:
- Number of active students today
- Number of inactive classrooms
- Backend and database health status

9. End-to-End Workflow

Professor uploads syllabus
↓
AI extracts curriculum
↓
Student learns topic
↓
Student practices questions
↓
Analytics engine updates
↓
Recommendations generated
↓
Revision queue updated
↓
Professor dashboard updated
↓
Admin dashboard updated

This creates a continuous feedback loop between learning, practice, analytics, and recommendations.